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Development of a broad bandwidth 193 nanometer laser driver for inertial confinement fusion

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ABSTRACT

The U.S. Naval Research Laboratory (NRL) has built a 137 J electron beam pumped argon fluoride (ArF) laser operating at 193 nm utilizing the Electra facility. This paper highlights initial ArF oscillator measurements in the context of significant benefits of ArF laser characterization for inertial confinement fusion target performance and development of a kinetics code with predictive capabilities. Other aspects of NRL's ArF laser development program, including large scale electron beam pumped ArF amplifier development and ArF optical transport, are discussed. Initial measurements were made utilizing a quadruple pass oscillator configuration constructed in the Electra facility with a nominal 100 cm^2 aperture. Laser yields between 80 and 100 J, time-dependent laser intensity, and amplified spontaneous emission (ASE) are reported over a range of gas pressure (0.8–1.4 atmospheres). Higher laser yields, 137 J, were attained in a non-optimized dual ArF oscillator configuration wherein the optical axis was rotated to conform to the measured electron beam deposition in the laser gas mixture.

1. Introduction

The U.S. Naval Research Laboratory (NRL) has built a 100 J class electron beam pumped argon fluoride (ArF) laser utilizing the Electra facility [1] in an effort to advance ArF laser technology. The ArF laser due to its short wavelength, 193 nm, and broad native bandwidth, 5 THz, has distinct advantages as a prospective driver for inertial confinement fusion. The shorter ultraviolet wavelength and broad bandwidth increases energy coupling to the target due to more efficient absorption and reduction in laser plasma instabilities. In addition, ArF has the potential of being the most efficient excimer laser from wall plug to lasing photons, [2,3] which reduces the size of the laser system and directly lowers the cost. The previously developed high energy excimer laser technologies [1] of induced spatial incoherence (ISI) beam smoothing [4,5] and flexible control of focal spot utilizing zooming [6] are straightforward with ArF lasers. The laser medium is a gas which is easy to cool and enables capability of many shots per day for the largest conceivable laser systems. The improved capability of a large scale ArF laser system compared to other potential inertial confinement fusion laser drivers is beneficial and would allow increased capability for both direct drive [7] and indirect drive [8,9] concepts. The modular nature to build duplicative systems of electron beam pumped excimer lasers for large scale systems along with the above mentioned attributes of the ArF laser including high efficiency, broad bandwidth and short wavelength make ArF lasers a desirable candidate for laser inertial confinement fusion driver.

In regard to the potential benefits of the ArF laser driver, NRL has embarked on an effort to investigate basic research and development of electron beam pumped ArF lasers. Previous research in high energy ArF laser measurements is limited: the largest ArF laser outputs prior to this work were 96 J at Keio University in Japan [10] and 92 Joules at Sandia National Laboratory [11]. NRL is using an integrated approach, which includes both modeling [12] and experiments [13] to validate and develop a predictive capability for the electron beam ArF laser driver. Such predictive capability will establish high confidence in designing and building large scale laser systems.

This paper reports a quadruple pass oscillator configuration that was constructed in the Electra facility with a 115 cm^2 aperture. Laser yield, time-dependent laser intensity, and amplified spontaneous emission (ASE) were measured over a range of laser gas pressures (0.8–1.4 atmospheres). This paper also reports preliminary results which have good agreement between ArF oscillator experiments and the NRL developed ArF kinetics code, Orestes. In addition, we report laser yield with a reconfiguration of the optical design to two double

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pass oscillators in parallel (dual parallel oscillators). Our previous work showed agreement between experiments and theoretical calculations for the ArF laser small signal gain as a function of power deposition [13]. Measurements of the small signal gain, non-saturable absorption, and saturation intensity of ArF as a function of laser gas pressure were made in an electron beam pumped amplifier experiments of both single pass and double pass configuration in a wide range of conditions. For example, at a pressure of 0.82 atm (12 psia), laser gas composition of 0.3% fluorine and 99.7% argon, and power deposition of 1.09 MW/cc the measurements yielded 3.21%/cm small signal gain, 0.16%/cm non-saturable absorption, and 10 MW/cm^2 saturation intensity [13]. In separate experiments, a gated intensified camera was used to study the time-resolved e-beam deposition in the laser gas as a function of gas pressure [13]. These measurements allow an evaluation of the spatial profile of the energy deposition in the gas as well as the power deposition per volume. The results of the small signal gain with the electron beam deposition profile suggest a high intrinsic efficiency of 17.7% for ArF lasers [13]. Data from the amplifier experiments as well as the oscillator measurements provide a rigorous evaluation test of the NRL-developed, excimer laser kinetics code Orestes. The program goals are to evaluate the laser performance over a range of conditions including pressure, e-beam deposition, and laser gas composition; advance the NRL Orestes code to be a reliable and predictive tool for designing large scale e-beam pumped ArF lasers for optimal output; and develop the required e-beam technologies to fabricate large scale ArF lasers for fusion applications.

2. Experiment

Electra has been recently configured as an oscillator to examine ArF physics, in order to determine how the various laser components work together and to evaluate the reactivity of the laser cell, hibachi-foil structure, and windows to fluorine and ArF laser light. Multiple optical configurations were used for preliminary evaluation of the ArF laser performance around the laser gain media generation volume. The laser gain media generation volume is enclosed by a pair of windows, Corning 7980 standard grade 1, inclusion class F with dimensions of 32.7 × 34.7 × 3.8 cm^3 separated by 135 cm. The measured transmission of the uncoated re-polished fused silica laser cell windows is 90%. The laser cell windows are tilted at angle of 14 degrees, each in a parallelogram configuration to eliminate parasitic buildup of amplified spontaneous emission (ASE). Parasitic buildup of ASE would deplete the laser efficiency due to destruction of ArF molecules formed by the deposited electron beam energy in the gas. The windows are sealed to a stainless steel laser cell, which has a total volume of 879 liters with numerous ports for diagnostic access. The laser cell is filled with an admixture of fluorine and argon, whose composition (percentage of fluorine up to 5%) and pressure from low 10^{-6} Torr (1.3×10^{-9} atmospheres) up to 1900 Torr (2.5 atmospheres) can be varied. The two counter-propagating electron beams enter the laser cell orthogonal to the laser optical axis through a 25.4 μm (1 mil) thick stainless steel foil, which is a barrier between the laser gas mixture and the vacuum diode. The foil is supported by a metal structure called a hibachi. The electron beams are generated by two pulsed power systems that generate 500 keV electrons of total current 110 kA in a profile having a flat-top of 100 ns and a pulse width of 140 ns. Since ArF laser performance is improved with higher electron beam deposition power [13], velvet cathodes of 10 × 100 cm^2 were utilized to attain peak power deposition close to 1 MW/cc. A velvet emitter was chosen due to the fast rise-time and spatially uniform electron beam emission.

A 1.4 kG external axial magnetic field oriented normal to the laser optical axis is applied to both vacuum diode boxes and laser cell which rotates the electron beam in the vacuum boxes relative to the initial cathode orientation by 1.6 degrees. The electron beam deposition volume is estimated to be 14 × 30 × 100 cm^3 from imaging measurements. [13] The total energy deposition into the laser cell was measured using a pressure transducer gauge with implementation of the ideal gas approximation for the dominant gas 99.7% argon and a correction for radiation losses (including laser and ASE) within the response time of pressure transducer. [14] Rogowski coils and capacitive probes measure time resolved diode current and voltage, respectively. The time dependent diode current and voltage, along with the minimum voltage energy to pass through the 25.4 μm (1 mil) foil is used to create a power pulse shape. The power pulse shape is utilized with the total energy deposition in the laser gas to find the time-dependent power deposition. Time dependent laser output and ASE are measured with 1 ns rise-time photodiodes fielded with 193 nm interference filters and attenuated with neutral density filters.

3. ArF kinetics code

The NRL code for excimer laser kinetics (Orestes) is under revision from KrF applications to model the gas kinetics and yield of the ArF excimer laser. The Orestes code combines electron kinetic treatment with non-equilibrium plasma chemistry (1D), lasing amplification (1D) and 3D ASE radiation transport. The first step is to solve the electron Boltzmann equation for the Electron Energy Distribution Function (EEDF) for a wide range of electron beam power depositions and laser compositions. The EEDF determines the electron kinetics which includes excitation, ionization, attachment, and all other kinetic rates directly related to electrons. The electron kinetics have been rigorously treated by numerically solving the steady-state, spatially averaged electron Boltzmann equation in Ar-F_2 plasma. [12] Another module of the code, beyond the EEDF, calculates the plasma chemistry which follows the formation and destruction of 16 species in 60 reactions within the ArF laser system (Fig. 1). The code tracks atomic and molecular species for direct time-dependent comparisons to measureable key quantities such as populations of ArF, Ar_2F, F_2, and electron density. Fig. 1 shows a schematic of the plasma chemistry and the species tracked by the ArF Orestes code. There are two channels for the upper laser level ArF* population. The neutral channel creates excited Ar atoms followed by harpoon reactions with fluorine. The ion channel has formation of argon ions (Ar^+) followed by ion-ion recombination utilizing F^-. The destruction mechanisms of ArF* are collisions with argon atoms (Ar), fluorine molecules (F_2), fluorine atoms (F), electrons (e^-), and radiation decay (γ) including stimulated emission of ArF*. In addition, the plasma kinetic module calculates populations of absorbing species taking into account absorption (α) of 193 nm in the gas medium. The combination plasma chemistry modules and radiation transport module provides a predictive capability for ArF electron beam pumped lasers.

4. Quadruple pass oscillator

A quadruple pass oscillator Fig. 2) was designed and constructed to evaluate the performance of ArF as a lasing medium. The quadruple pass oscillator was constructed to extend the lasing path and compensate for the low small signal gain in ArF as well as measure the saturation intensity and efficiency of a possible ArF laser amplifier. The uncoated fused silica output coupler is made of Corning 7980 ArF grade 0F material with dimensions of 14.0 × 14.0 × 3.2 cm^3 with a clear aperture of 11.9 × 11.9 cm^2. The reflectivity of the uncoated fused silica output coupler was nominally 8% in measurements with 600 mJ LPX 305i ArF laser. The rest of the cavity is composed of high reflectors with a reflectivity of 97%. The zero degree high reflector dimensions of 14 × 14 cm^2 with a clear aperture of 11.9 × 11.9 cm^2 was aligned to be parallel to the output coupler. The two 14 × 19 cm^2 45 degree mirrors completed the four-optic cavity. The results of utilizing the four-optic cavity are shown in Table 1 with a pressure of 1.09 atm (16 psi) and a laser gas mixture of 99.7% Ar and 0.3% F_2, which produced the highest laser yield of 96 J for this configuration. The average power deposition for the gain medium volume was 0.8 MW/cc. The value of

species:	energy (eV):
Ar	0.00
Ar^*	11.65
Ar^{**}	13.17
Ar_2	0.00
Ar_2^*	10.60
F	0.82
F^*	12.7
F_2	0.00
F_2^*	11.57
ArF^*	8.30
Ar_2F^*	6.00
Ar^+	15.76
F_2^+	15.69
Ar_2^+	14.53
F^-	-2.58
electrons	

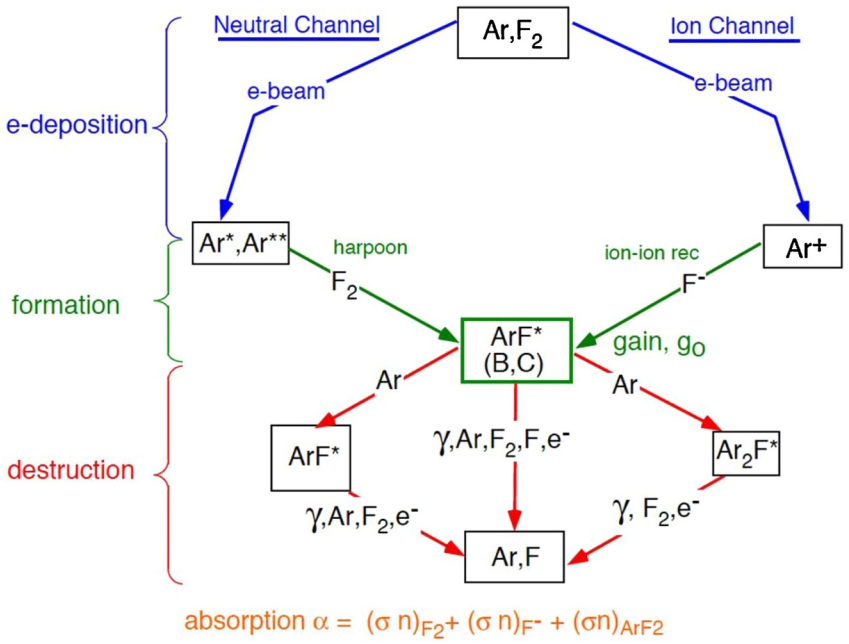


Fig. 1. NRL ArF Orestes code schematic (right) of electron energy deposition into initial laser gas mixture, formation and destruction mechanisms of ArF* with a table of species utilized (left).

0.8 MW/cc is somewhat less than the 1 MW/cc utilized in the previous small signal gain measurements due to averaging over a larger volume and having some non-uniformity in the electron beam deposition. Therefore we expect that increased uniform electron beam power deposition throughout the volume would increase the laser yield. At higher pressures of 1.36 atm we observed a decline in laser yield (80 J). Presumably this decline is due to kinetic effects such as 3-body dissociation reactions of ArF, namely the reactions (1).

$$ArF + Ar + Ar \rightarrow Ar_2 + Ar + F \text{ or } Ar_2F + Ar \quad (1)$$

Lowering the pressure to 0.82 atm with 99.7% Ar and 0.3% F_2 composition had a slight reduction of laser yield to 90 J, but demonstrated a higher intrinsic efficiency than the highest yield condition of 1.09 atm (Table 1). The intrinsic efficiency was estimated from the maximum laser power divided by the average electron beam power deposition for the last pass through the quadruple pass oscillator. The laser pulse shape, Fig. 3, was utilized to find the peak laser power in MW/cm^2 along with a transmission correction for the final window (89%) and output coupler (90%). The estimated stage gain for the final pass was 10, therefore laser input was subtracted from the laser output of 10% to account for the stage gain. The estimate for intrinsic efficiency has a significant error due to the estimate of the peak laser intensity, optical transmission as well as the electron beam power deposition. An important point is that an oscillator configuration may not be able to extract all the possible laser energy for a given electron beam energy deposition due to the lack of control over optical rays and the additional ASE flux allowed to pass around the gain media without suppression.

The best agreement between ArF oscillator experiments (0.82 J/cm^2 over 10.5 × 11 cm^2 laser aperture) and NRL Orestes kinetics code (0.84 J/cm^2) occurred at a total pressure of 0.82 atm, with a fraction of 0.3% F_2 in the argon laser mixture using a power deposition of 0.6 MW/

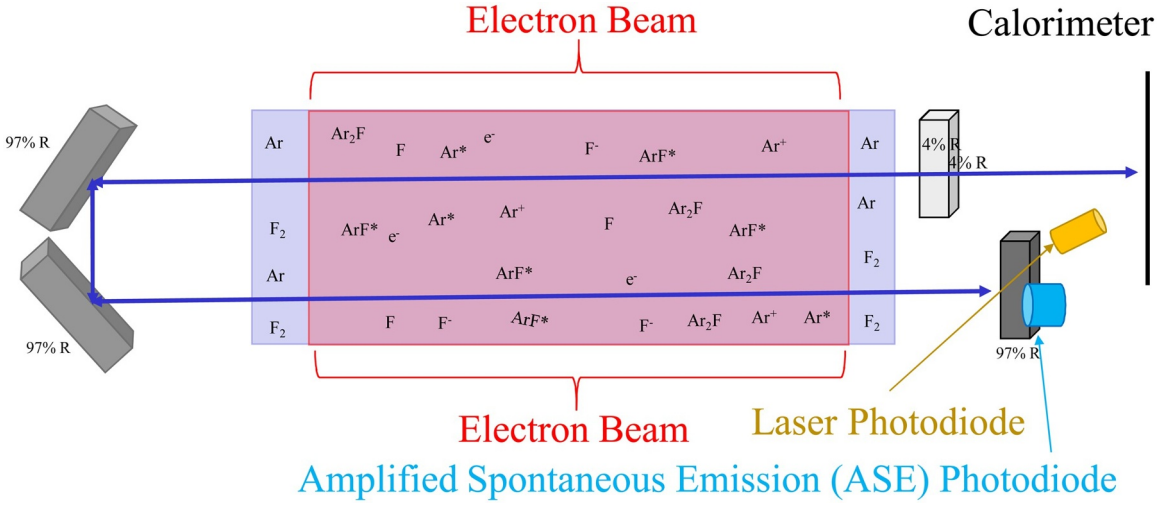


Fig. 2. Top down view of quadruple pass ArF oscillator. Electron beam is injected from top and bottom (in figure) with a configuration of output coupler, two 45 degree high reflectors and one 0 degree high reflector. Diagnostic positions of calorimeter and photodiodes are also shown. The amplified spontaneous emission photodiode was above the zero degree in the cavity looking at an angle into the laser gain medium.

Table 1
Performance of quadruple pass ArF oscillator as a function of pressure, fractional composition and electron beam energy deposition. The highest yield was measured at 1.09 atm with 96 J. Estimated peak power efficiency, evaluated as peak laser power divided by peak electron beam power, was included even though the attainment of optimal conditions was not reached and measurements were conducted in an oscillator mode which is not the same as in amplifier configuration.

Laser energy	Pressure	Composition	Flat-top peak power	Estimated peak power efficiency
80 J	1.36 atm (20 psia)	99.7% Ar, 0.3% F_2	0.9 MW/cc	12%
96 J	1.09 atm (16 psia)	99.7% Ar, 0.3% F_2	0.8 MW/cc	14%
90 J	0.82 atm (12 psia)	99.7% Ar, 0.3% F_2	0.6 MW/cc	16%

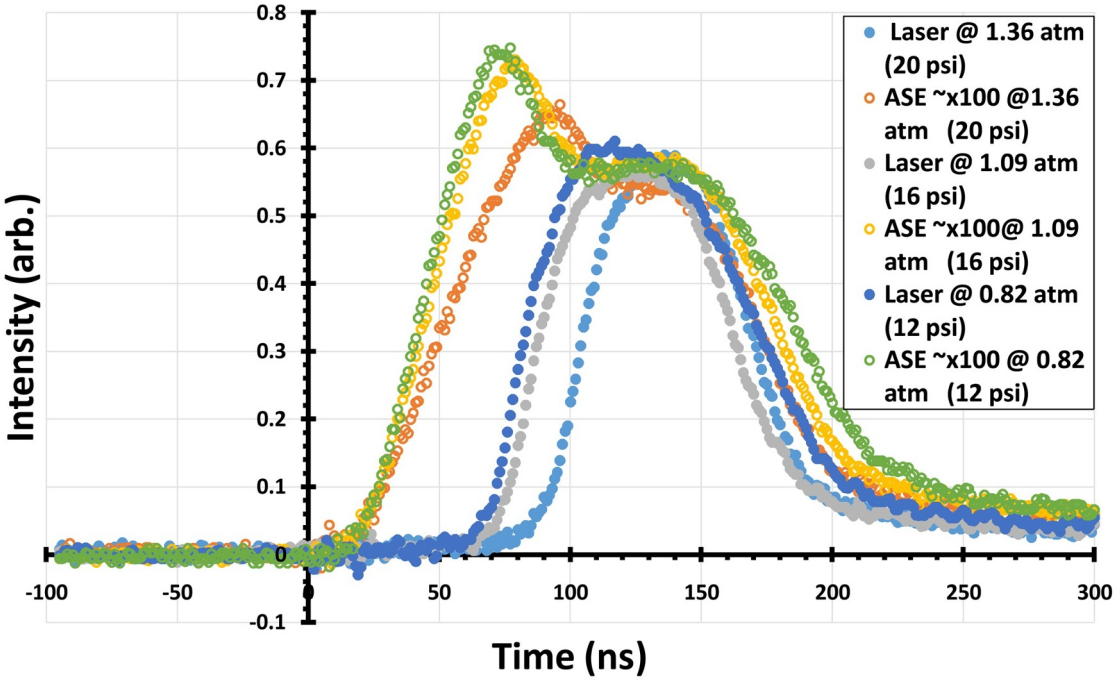


Fig. 3. Time-dependent measurement of output laser and amplified spontaneous emission (ASE) as a function of pressure for the quadruple pass ArF oscillator. The ASE was multiplied by 100x to allow visual comparisons of ASE and laser output.

cc. The 0.82 atm pressure condition is the maximum intrinsic efficiency condition for these measurements and is slightly lower than maximum yield. The higher pressure allows greater stopping of the electron beam power and therefore, the intrinsic efficiency, which is laser power divided by electron beam power deposition, decreases due to the increase in electron beam power and modest decrease of laser energy. However, the best agreement did occur at the highest intrinsic efficiency conditions for which the code has been developed to find. The 10.5 × 11 cm^2 aperture was the measured beam size during the experiments, which was less than the maximum 11.9 × 11.9 cm^2 laser beam aperture that could be attained. The discrepancy between the aperture size is surmised to be misalignment (offset positioning of 45 degree mirrors relative to zero degree mirror and output coupler), but further investigation is underway. At higher pressures the Orestes code predicts significantly higher laser yield than the experiments. The discrepancy is attributed to small errors in reaction rates for 3-body reactions as in Eq. (1).

Measurements of the time-dependent laser intensity and amplified spontaneous emission were made as a function of pressure. The ASE, which has been enhanced by ~100x in Fig. 3 initially rises before the onset of the laser due to the low output coupler reflectivity. The ASE was measured from a position above the zero degree mirror utilizing the 32.7 × 34.7 cm^2 windows that were much larger than the gain media. This allows an examination of the diagonal region through the gain media to maximize length and signal response. Many roundtrip passes are required to get sufficient gain in the laser pulse to measure a signal with the ArF photodiode under sufficient attenuation to prevent photodiode signal saturation. As the laser output rises suppression of ASE is observed initially at 70 ns and 75 ns for the 0.82 atm and 1.09 atm, respectively. The initial suppression of ASE for 1.36 atm does not occur until 90 ns. The delay in all three cases of ASE suppression shows the laser extraction could be better optimized with greater reflectivity and/or smaller cavity length. The higher pressure photodiode response for ASE and lasing output starts to decline earlier in time (~130 ns in Fig. 3) due to the increased presence of 3- body interactions destroying ArF molecules due to the higher densities of molecules, ions and atoms. The laser energy measured from the time integrated time-dependent photodiode measurement is higher for 0.82 atm than 1.09 atm which is different than the data measured by the calorimeter in Table 1. The time-dependent laser photodiode did not encompass the entire laser beam aperture and shows some differences than the calorimeter measurements as a function of pressure. The differences can be accounted for due to changes in the laser beam uniformity as a function of pressure. We are aware of non-uniformity in our counter-propagating electron beam source which are most prominent at low pressures [13]. A properly designed ArF laser amplifier would extract high laser intensity throughout the entire high intensity ASE temporal range (60–160 ns in Fig. 3). Therefore amplifier configurations are expected to have much higher laser efficiency and laser yield than in this oscillator configuration.

5. Dual parallel oscillators

The optical configuration was modified (Fig. 4) into a dual parallel

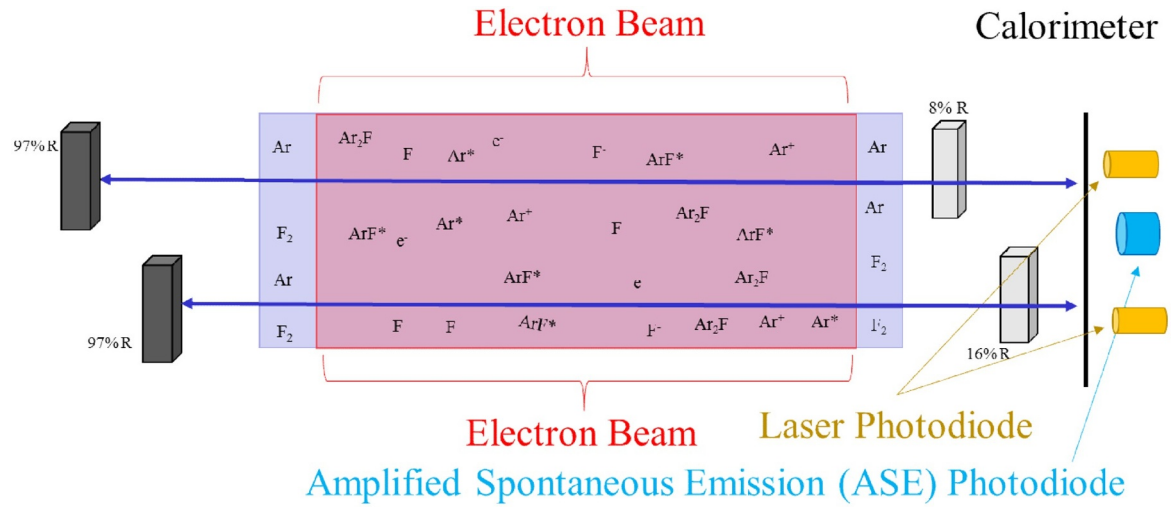


Fig. 4. Top down view of dual parallel ArF oscillator. Electron beam is injected from top and bottom (in figure) with a configuration of 8% output coupler opposing 0 degree high reflector parallel to second oscillator composing of a 16% output coupler opposing 0 degree high reflector. Diagnostic positions of calorimeter and photodiodes are also shown. The amplified spontaneous emission (ASE) was measured from above the calorimeter examining a diagonal volume of the laser gain medium.

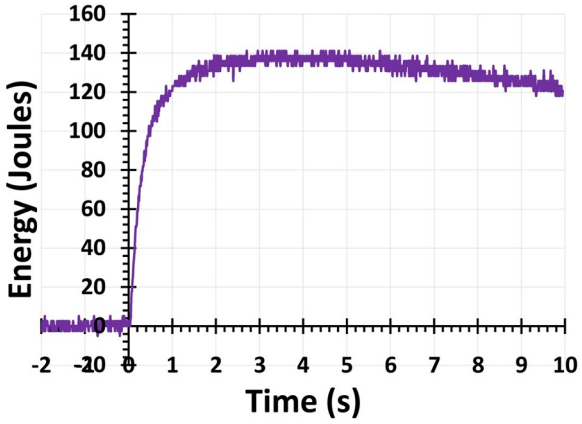


Fig. 5. Calorimeter voltage response converted to energy (J). ArF laser yield of 137 J measured utilizing dual parallel oscillators with laser mixture of 99.71% Ar and 0.29% F_2 encompassing the output of both laser beams simultaneously from an instantaneous 100 ns laser pulse.

non-optimized ArF oscillator system. The dual parallel oscillator was configured to garner greater laser yield by increasing the effective aperture with the known detriment of operating significantly below the saturation intensity and at a lower efficiency. Therefore ArF laser evaluation could take place over a great range of conditions with the present electron beam power system and optics. Compared to the quadruple pass oscillator, the modifications increased the laser yield to 137 J. The modification consisted of utilizing two high reflectors with 11.9 × 11.9 cm^2 clear apertures directly opposing through the laser gain media with output couplers of 8% and 16%. The 8% output coupler is the aforementioned output coupler with ArF grade 0F fused silica uncoated. The 16% output coupler is a 193 nm partial reflector of 16% utilizing ArF grade 0F of dimensions 14.0 × 14.0 × 3.2 cm^3 with an anti-reflection coating (<1%) at 193 nm on the polished second surface. Both oscillator laser beams are intercepted with the 33 × 33 cm^2 calorimeter (Fig. 5) to measure a total of 137 J. The conditions, used to get 137 J, were 99.71% Ar, 0.29% F_2 at 0.95 atmospheres (14 psi). The result was not optimized for fluorine concentration or pressure, but was close to optimal conditions found in the quadruple pass oscillator measurements. In addition, the dual oscillators utilize two different output couplers and we have not completely elucidated the relative performance of these two oscillators with output coupler reflectivity. Laser energy measurements were carried out with a calibrated 33 × 33 cm^2 calorimeter that is robust to ArF laser damage. We expect in the future to evaluate each oscillator separately once calorimeters of appropriate dimensions are acquired. Optimization of output coupler reflectivity could enhance the ArF laser output.

6. Conclusions and future work

In this paper we have summarized our initial ArF oscillator results on the Electra laser facility at NRL. We have attained 137 J in a non-optimized parallel dual oscillator configuration. NRL is in the process of advancing ArF laser technology by developing Electra to reach higher laser (~1 kJ) energies. Throughout all the laser runs, we have not incurred any transmission loss at our operating fluences. Commercial vendors' state 193 nm reflector laser induced damage thresholds of 2 J/cm^2. Research on AlF_3/LaF_3 high reflector coatings has shown higher laser damage thresholds (4.5 J/cm^2 at 193 nm). [15] Window materials such as CaF_2 exceed 6 J/cm^2 for laser induced damage threshold at 193 nm. Based on our initial ArF laser performance, NRL is interested in evaluating the advancement of optical damage thresholds at 193 nm for which it has built a Class 1000 testing environment. NRL is also examining concepts for development of larger scale (10's kJ–100 kJ) ArF laser systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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